Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/17/2005
PI Name:	Lelkes, Peter Istvan Ph.D.		
Project Title:	Altered Signal Transduction and Differential Gene Expression in PC12 Pheochromocytoma Cells Cultured in RWV Bioreactors		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	pilelkes@temple.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	215-204-3307
Organization Name:	Temple University		
PI Address 1:	Department of Bioengineering		
PI Address 2:	1947 N 12th St, Room 811		
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19122-6018	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-02
Start Date:	09/01/2000	**End Date:**	05/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1436		
Performance Goal No.:			
Performance Goal Text:			
	The focus of this ground-based proposal is to characterize signaling mechanisms and differential gene expression in PC12 pheochromocytoma cells cultured in NASA's Rotating Wall Vessel (RWV) Bioreactors, a low-shear, microgravity (microG)- simulating cell culture venue. PC12 cells, neural crest derived and bipotential, are well established cellular models for studying signal transduction involved in neuronal/neuroendocrine differentiation Preliminary data suggest alterations in cellular signaling patterns and differential gene expression in PC12 cell maintained in RWV Bioreactors for only short periods of time (hours). Based on these observations it is hypothesized that the microgravity-simulating cell culture environment in NASA's RWV Bioreactors is sufficient to induced differential gene expression through alterations in phosphorylation-dependent signaling pathways. The overall objective of this study is to test the hypothesis that that the microgravity-simulating cell culture environment in NASA's RWV Bioreactors is sufficient to induced differential gene expression through alterations in phosphorylation-dependent		

Task Description:	signaling pathways. The following specific aims are proposed to test this overall objective: 1. Study the time course of the acquisition of a neuroendocrine/adrenergic phenotype in PC12 pheochromocytoma cells cultured in simulated microgravity. 2: Study the time course of the down-regulation of the neuronal phenotype in PC12 cells cultured in simulated microgravity. 3: To investigate the cellular and molecular mechanisms by which simulated microgravity alters protein phosphorylation dependent signaling pathways. 4: Establish a causal link between simulated microgravity-induced alteration in cellular signaling and PNMT activation. Of immediate significance, the results of this study will provide an understanding of specifically altered signaling pathways, which are involved in the unique differentiative capabilities of RWV Bioreactors. While this study is carried out with one well characterized cell line, viz PC12 pheochromcytoma cells, it is anticipated that the outcome of this study will be widely applicable to other cellular systems, as well. Of far reaching significance is the long-term goal, namely to provide NASA with a well-characterized system, responsive to simulated microG, which will be ready for further testing on board the International Space Station.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study will contribute to the characterization and elaboration of NASA-designed RWV Bioreactors as a unique, microgravity-simulating cell culture environment here on earth. Also, this unique "in vitro embryology" venue will help to identify unique signaling pathways involved in the crucial steps necessary for the formation of differentiated tissue-like assemblies from single cells and cell aggreates.
Task Progress:	In this past year we have achieved significant progress in all specific aims. In further refining our understanding of the time course for the acquisition of a neuroendocrine/adrenergic phenotype in PC12 pheochromocytoma cells cultured in simulated microgravity, we now corroborated and solidified last year's findings that some of the specific signaling pathways, necessary and sufficient to induce the neuroendocrine /adrenergic phenotype, are upregulated within the first two hours upon culture in the RWV environment. Furthermore, we confirmed that second messenger systems (cAMP), signaling through mitogen activated kinases and specific nuclear transcription factors, such as egr-1, GRE and CREB are rapidly upregulated in RWV. By contrast, other signaling pathways related to activation by the low-levels of fluid shear stress in the RWV Bioreactors are upregulated only at significantly later time points. For example, the nuclear transcription factor NF-kappa B is not activated for the first 6 hours. In continuation of last year’s experiments, we have expanded our analysis of the altered pattern of signaling in RWV Bioreactors using modern genomics, proteomics and signalomics approaches. Specifically, we are using gene expression profiling as well as global protein expression analysis using 2-D gel electrophoresis. We have finished these sets of experiments and are now in the process of evaluating the results. Given the large scope of these expression profiles (we can reproducible resolve approx 10,000 genes and some 1,500 proteins) , we are currently in the process of analyzing these data. Specifically, we are using modern techniques of bioinformatics and array biostatistics to mine our data bases in order to obtain meaningful correlations and to be able to establish clustering of signaling molecules. As previously reported we have been studying of the cellular and molecular mechanisms by which the modeled microgravity environment in RWV Bioreactors alters protein phosphorylation-dependent signaling pathways. As previously, we are using established inhibitors and activators which stimulate and abrogate some of these pathways. In extension of our previous work, we have this past year begun to assess not only individual phosphorylation dependent pathways, such as the MAPK pathays or the pathways involving PI3 Kinase. Rather, we have begun to look at global expression profiling of phosphorylated proteins and of significant number of protein kinases, using modern kinase array technology, and 2-D phosphoprotein gel electrophoresis. As another innovation, we are evaluating the data using state-of-the-art systems biology approaches and data mining tools , at the levels of genomics, proteomics and signalomics. The results form these studies form the basis for two publications which are currently under review. Taken together, this year’s experiments and results expand out previous data which helped to establish a causal link between simulated microgravity-induced alteration in cellular signaling and PNMT activation. Furthermore our studies a set the stage for assembling the experimental and analytical tools necessary for our long-term goal, namely to provide NASA with a well-characterized cellular system, responsive to simulated microgravity, which is now ready for further testing on board the International Space Station.
Bibliography Type:	Description: (Last Updated: 09/28/2023)
Abstracts for Journals and Proceedings	Troop EM, Jain H, Rezvan A, Somiari RI, Lelkes PI. "Specific Impact of Forces Present in Rotating Wall Vessel Bioreactors As Assessed by Gene Profiling." BMES 2004, Biomedical Engineering Society Annual Fall Meeting, Philadelphia, PA, October 13-16, 2004. BMES 2004, Biomedical Engineering Society Annual Fall Meeting, Philadelphia, PA, October 13-16, 2004. Abstract # 830. , Oct-2004
Awards	Lelkes PI. "Distinguished Guest Professor, Chinese Academy of Sciences, December 2004." Dec-2004
Books/Book Chapters	Lelkes PI, Unsworth BR, Saporta S, Cameron DF, Gallo G. "Culture of Neurendocrine and Neuronal Cells for Tissue Engineering." in "Culture of Cells for Tissue Engineering." Ed. G. Vunjak-Novakovic, R.I. Freshney. Hoboken, N.J. : John Wiley & Sons, 2006. p. 375-416., Jan-2006
Presentation	Lelkes, P.I. "Neuroendocrine Organogenesis using RWV Bioreactors" Symposium in Honor of B.R. Unsworth, Marquette University, Milwaukee, WI, August 20, 2004 Aug-2004